

2025 HSC Biology Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	A
4	B
5	B
6	D
7	C
8	D
9	A
10	C
11	C
12	D
13	A
14	A
15	C
16	A
17	C
18	B
19	C
20	D

Section II

Question 21

Criteria	Marks
• Identifies the role of two components of the innate immune system	2
• Provides some relevant information	1

Sample answer:

Component	How it protects against infection
Stomach acid	The acid inhibits the growth of or kills bacteria/pathogens
Skin	Acts as a physical barrier to pathogen entry into the body

Answer could include:

Nasal hair, tear glands, mucus lining, urinary tract

Question 22 (a)

Criteria	Marks
• Outlines a process used by fungi for reproduction	2
• Provides some relevant information	1

Sample answer:

Fungi can reproduce asexually by producing spores which are released and when carried into a favourable environment they germinate and grow into a new fungal organism.

Answer could include:

Budding

Question 22 (b)

Criteria	Marks
• Outlines how a pathogen adaptation facilitates transmission between hosts	2
• Provides some relevant information	1

Sample answer:

After colonising a host, fungi can produce spores which can be released to infect a new host.

Question 23

Criteria	Marks
• Compares artificial insemination and artificial pollination	3
• Provides an incomplete comparison of artificial insemination and artificial pollination	2
• Provides some relevant information	1

Sample answer:

A similarity is that both processes involve gametes of an organism. A difference is that artificial insemination occurs only in animals while artificial pollination occurs only in (flowering) plants.

Question 24 (a)

Criteria	Marks
• Provides correct example for both mechanism <i>A</i> and mechanism <i>B</i>	2
• Provides some relevant information	1

Sample answer:

Mechanism *A* – sweating
 Mechanism *B* – shivering

Question 24 (b)

Criteria	Marks
• Outlines how mechanism <i>B</i> maintains homeostasis	2
• Provides some relevant information	1

Sample answer:

When the body temperature drops, skeletal muscles start shivering. This generates body heat and raises the temperature. This helps maintain homeostasis.

Question 25

Criteria	Marks
- Provides an accurate evaluation of the product statement - Demonstrates a sound understanding of the data provided	3
Demonstrates some understanding of the product statement and the data provided	2
Provides some relevant information	1

Sample answer:

The product statement is incorrect because a person is not fully protected from UV. UV levels greater than 3 occur between approximately 8 am and 6 pm and so for the given day there is about 2 hours before and after recommended times, a person would have no protection from UV.

Question 26

Criteria	Marks
Provides a thorough comparison between LASIK and one other technology	4
Provides a sound comparison between LASIK and one other technology	3
- Provides some comparison between LASIK and one other technology OR - Demonstrates some understanding of LASIK OR another relevant technology	2
Provides some relevant information	1

Sample answer:

Named visual disorder – Myopia

Myopia can be corrected by wearing glasses with concave lenses which refracts the light so that the image falls on the retina rather than before it. This is similar to LASIK surgery where the cornea is reshaped to precisely refract the light to reach the retina at the back of the eye. LASIK is an intrusive and more expensive procedure whereas prescription glasses are less expensive and non-intrusive.

Question 27

Criteria	Marks
• Correctly identifies the type of fertilisation in each group • Provides an extensive comparison of internal and external fertilisation • Incorporates references to the data provided	6
• Correctly identifies the type of fertilisation in each group • Provides a thorough comparison of internal and external fertilisation • Makes some reference to the data provided	5
• Correctly identifies the type of fertilisation in each group • Demonstrates a sound understanding of internal and external fertilisation • Makes limited reference to the data provided	4
• Outlines feature(s) of fertilisation	2–3
• Provides some relevant information	1

Sample answer:

The animals in Group A are all terrestrial animals with internal fertilisation. The animals in Group B are all aquatic animals with external fertilisation.

In both groups, fertilisation involves fusion of male and female gametes.

In Group A this occurs inside the female body. Fewer gametes are needed as there is a higher probability of fertilisation in the internal space. Thus the animals in Group A produce far fewer eggs (eg 1 to 100 in snakes) than in Group B (100 000 to 2 million in sea urchins) and they are able to reproduce in terrestrial environments as internal fertilisation protects the gametes.

In Group B, fertilisation must occur in an aquatic environment so that the gametes do not dry out. Fertilisation in Group B has a much lower probability of success as the gametes are spread over a larger volume. Internal fertilisation ensure higher rates of survival of the fewer offspring produced.

Question 28 (a)

Criteria	Marks
• Correctly describes the process of antibody production with correct reference to the role of X, Y and Z	4
• Outlines the process of antibody production with some reference to the flow chart	3
• Outlines a process of antibody production	2
• Provides some relevant information	1

Sample answer:

The tick bite transfers antigenic alpha-gal sugar molecules into the skin. Macrophages, together with Helper T cells (X), activate B-lymphocytes (Y). These cells grow and differentiate into either memory cells, when exposure occurs, or plasma cells (Z) which produce antibodies against the antigen.

Question 28 (b)

Criteria	Marks
• Describes features of antibody production shown in the graph	2
• Provides some relevant information	1

Sample answer:

On first exposure to the alpha-gal sugar by the tick bite, antibody production takes longer and only a lower concentration of antibodies is produced. On re-exposure, antibodies are produced more quickly and a higher concentration is produced.

Question 28 (c)

Criteria	Marks
• Explains the role of memory cells in the immune response	3
• Outlines the role of memory cells	2
• Provides some relevant information	1

Sample answer:

Memory cells are created by B-cells in response to a specific antigen. Memory cells remain in the lymph nodes after the infection. On re-exposure to the antigen, these memory cells quickly differentiate into plasma cells or cytotoxic T-cells, which neutralise the pathogen more rapidly to prevent disease.

Question 29 (a)

Criteria	Marks
• Provides justification for <i>Varroa</i> mite infection being an infectious disease	2
• Provides some relevant information	1

Sample answer:

Varroa mite is an infectious disease because the mite is a parasitic pathogen that is spread between bees and hives that come into direct contact with the pest.

Question 29 (b)

Criteria	Marks
• Explains TWO procedures used to prevent the spread of <i>Varroa</i> mite	4
• Explains ONE procedure OR • Describes procedures used to prevent the spread of <i>Varroa</i> mite	3
• Describes ONE procedure OR • Outlines TWO procedures	2
• Provides some relevant information	1

Sample answer:

Measures used to prevent the spread of *Varroa* mite include:

- Early identification of *Varroa* mite infestation ensures early detection and isolation of infected bee populations. This will stop them from spreading to healthy bees
- Destroy infected hives. This will kill the mites and infected bees, preventing the spread of the mite to healthy hives.

Answers could include:

- Regular monitoring of bee colonies
- Minimising exposure by washing equipment and disinfection

Question 30 (a)

Criteria	Marks
• Describes an appropriate process used to produce genetically modified mosquitoes	4
• Outlines an appropriate process used to produce genetically modified mosquitoes	3
• Shows some understanding of how genetically modified mosquitoes are produced	2
• Provides some relevant information	1

Sample answer:

To produce the genetically modified (GM) mosquitoes, it is necessary to transfer the SERPINE 1 gene from a human to the mosquito using recombinant DNA technology. Restriction enzymes are used to cut the SERPINE 1 gene from a human cell chromosome. A bacterial plasmid (circular DNA) is opened using the same restriction enzymes and the SERPINE 1 gene is inserted and attached using DNA ligase. The bacteria then reproduce, producing multiple copies of the SERPINE 1 gene. The gene can then be delivered into the mosquitos. This can be done by micro-injection into mosquito egg cells, so that the mosquito which develop from the egg will contain the SERPINE 1 gene, which will allow the mosquito to express PAI-1.

Question 30 (b)

Criteria	Marks
• Demonstrates an extensive understanding of the impacts of genetic technologies for society • Provides a relevant judgement	7
• Demonstrates a thorough understanding of the impacts of genetic technologies for society • Provides a judgement	5–6
• Demonstrates a sound understanding of the impacts of genetic technologie/s for society	3–4
• Demonstrates some understanding of the impact of a genetic technology	2
• Provides some relevant information	1

Sample answer:

In recombinant DNA technology, selected genes can be cut and pasted from one organism into another using, for example, ligase enzymes. The human insulin producing gene has been transferred into bacteria, allowing for increased production of insulin, benefitting sufferers of diabetes, who can access insulin more readily and cheaply. This means that diabetes patients have a greater life span.

Genetic technologies such as CRISPR can be used for controlling pest populations including insects that cause diseases in both humans and other animals. By controlling the number of disease causing insects eg mosquitoes that cause dengue fever, we can significantly reduce the incidence of the vector-borne diseases. This would result in healthier populations and reduced stress on the health care system.

These technologies have largely been beneficial to society, improving access to drugs, the life expectancy and quality of life for many people.

Answers could include:

Benefits of the following genetic technologies

- Artificial insemination
- Artificial pollination
- Cloning.

Argument against genetic technologies

- Unintended health effects
- Environmental risks
- Ethical concerns.

Question 31 (a)

Criteria	Marks
- Identifies the type of inheritance from the pedigree - Provides an appropriate justification	3
Demonstrates some understanding of the type of inheritance from the pedigree	2
Provides some relevant information	1

Sample answer:

Type of inheritance: autosomal recessive

	b	b
B	Bb	Bb
b	bb	bb

BB – unaffected

bb – affected

Bb – unaffected

Individuals 3 and 4 have affected and unaffected children. Since 9 is an unaffected male, 3 would have to be heterozygous with genotype Bb and 4 would be homozygous recessive. For 10 to be affected, they would have inherited the recessive allele from the father and the mother.

Answers could include:

Individuals 8 and 9 are unaffected but have an affected child (17) thus the disease is recessive and was inherited from both parents making it autosomal. Individual 15 would be affected if it was sex-linked.

Question 31 (b)

Criteria	Marks
- Explains the type of mutation - Makes reference to the data provided	3
Outlines the mutation with reference to the data provided	2
Provides some relevant information	1

Sample answer:

The amino acid sequence in the unaffected individual is Glutamine-Tyrosine-Valine-Aspartic acid. The codon GUC coding for Valine is replaced with AUC, producing Isoleucine. As a single nucleotide is changed, CAMT is a point mutation.

Answers could include:

Missense, substitution

Question 32

Criteria	Marks
• Designs an appropriate epidemiological study that could produce most features of the graphed results • Justifies the features/steps with reference to the information provided	7
• Designs an appropriate epidemiological study that could produce some features of the graphed results • Justifies feature(s)/step(s) with reference to the information provided	5–6
• Provides some features of an epidemiological study • Makes some reference to the information provided	3–4
• Demonstrates some understanding of epidemiological studies	2
• Provides some relevant information	1

Sample answer:

Individuals from age groups 10, 20 and 30 in areas A, B and C should be surveyed over 55 years as shown in the graph. Areas A and B are connected via a road bridge so the communities can easily mix but area C is isolated so would serve as a control group.

The study should include equal numbers of males and females to ensure validity. To determine the risk of developing the disease due to age at exposure, data must be collected on other relevant features including diet, exercise, exposure to disease causing agents, general lifestyle and daily activities, and health of individuals.

Correlating the data to find common features or activities would help determine the risk of disease due to age at exposure. Surveying large numbers of individuals would improve the validity of the study.

Question 33 (a)

Criteria	Marks
• Compares the cell division processes in cell <i>R</i> and cell <i>S</i>	3
• Describes a cell division process	2
• Provides some relevant information	1

Sample answer:

Cell *R* undergoes mitosis which results in two genetically identical daughter cells. Cell *S* undergoes meiosis with half the chromosome number producing gametes. Both mitosis and meiosis will require DNA replication where the genetic content doubles.

Question 33 (b)

Criteria	Marks
• Correctly explains the relationship between the two individuals	2
• Provides some relevant information	1

Sample answer:

Individual 1 is the parent of Individual 2 because they provide a gamete to produce the offspring which is Individual 2.

Question 33 (c)

Criteria	Marks
• Provides a thorough analysis of how the separate mutations account for genetic differences in the cells	4
• Provides a sound analysis of how the separate mutations account for genetic differences in the cells	3
• Demonstrates some understanding of how the separate mutations account for genetic differences in the cells OR • Provides a sound analysis of a mutation	2
• Provides some relevant information	1

Sample answer:

Mutation *A* is a germline mutation and will not be present in Individual 1, but could be present in its offspring, including individual 2. Mutation *B* is a somatic cell mutation and will only be present in some cells of Individual 2 but not their offspring. As a result, cell *U* will not have mutations *A* or *B*. Cell *V* could have mutation *A* but not *B* and cell *W* could have both mutations. Cell *X* could carry mutation *A*.

Question 34 (a)

Criteria	Marks
• Explains why the small population allele frequency fluctuations are more pronounced compared to that of a larger population	2
• Makes a comparison between the two groups	1

Sample answer:

The smaller the population size, the smaller the gene pool. Small gene pools have a decreased genetic diversity, so any fluctuations in allele frequency will have a greater effect on the gene pool.

Question 34 (b)

Criteria	Marks
• Evaluates the effects of gene flow on a gene pool for both small and large populations	4
• Describes the effects of gene flow on a gene pool for both small and large populations	3
• Describes an effect of gene flow on a gene pool for either small or large populations	2
• Provides some relevant information	1

Sample answer:

Small populations have limited genetic diversity. Gene flow can benefit the gene pool by introducing genetic variation into a population. If the environmental conditions are favourable, the gene can easily spread to increase the population's ability to adapt to a changing environment. Where the environmental conditions are unfavourable, the gene will quickly be removed due to natural selection, as shown in the graph of the small population where the frequency of the introduced allele is shown to dramatically change in frequency.

The impact of the introduced gene is less pronounced in larger populations, due to the higher level of genetic variation already present. Gene flow generally has a positive effect by further increasing genetic diversity. The graph shows that the allele frequencies of a larger population fluctuate little over time as the introduced allele is of a lower proportion in the gene pool.

The impact of gene flow can be positive or negative depending on population size and environmental conditions.

2025 HSC Biology Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	Mod 5 Reproduction	12-12
2	1	Mod 6 Mutation	12-13
3	1	Mod 7 Immunity	12-14
4	1	Mod 6 Mutation	12-13
5	1	Mod 7 Causes of Infectious Disease	12-14
6	1	Mod 7 Prevention, Treatment and Control	12-14
7	1	Mod 8 Homeostasis	12-5, 12-15
8	1	Mod 5 Genetic Variation	12-12
9	1	Mod 8 Technologies and Disorders	12-6, 12-15
10	1	Mod 7 Prevention, Treatment and Control	12-5, 12-14
11	1	Mod 7 Prevention, Treatment and Control	12-14
12	1	Mod 6 Mutation	12-13
13	1	Mod 8 Epidemiology	12-6, 12-15
14	1	Mod 7 Immunity	12-14
15	1	Mod 8 Technologies and Disorders	12-6, 12-15
16	1	Mod 6 Genetic Technologies	12-13
17	1	Mod 5 Cell Replication	12-6, 12-12
18	1	Mod 5 Genetic Variation	12-6, 12-12
19	1	Mod 5 Cell Replication	12-6, 12-12
20	1	Mod 8 Causes and Effects	12-5, 12-13, 12-15

Section II

Question	Marks	Content	Syllabus outcomes
21	2	Mod 7 Immunity	12-14
22 (a)	2	Mod 5 Reproduction	12-12
22 (b)	2	Mod 7 Causes of Infectious Disease	12-14
23	3	Mod 6 Genetic Technologies	12-13
24 (a)	2	Mod 8 Homeostasis	12-15
24 (b)	2	Mod 8 Homeostasis	12-6, 12-15
25	3	Mod 8 Prevention	12-5, 12-15
26	4	Mod 8 Technologies and Disorders	12-15
27	6	Mod 5 Reproduction	12-5, 12-12
28 (a)	4	Mod 7 Immunity	12-5, 12-14
28 (b)	2	Mod 7 Immunity	12-5, 12-14
28 (c)	3	Mod 7 Immunity	12-5, 12-14

Question	Marks	Content	Syllabus outcomes
29 (a)	2	Mod 7 Causes of Infectious Disease	12-14
29 (b)	4	Mod 7 Prevention, Treatment and Control	12-14
30 (a)	4	Mod 6 Genetic Technologies	12-13
30 (b)	7	Mod 6 Genetic Technologies	12-6, 12-13
31 (a)	3	Mod 5 Genetic Variation	12-6, 12-12
31 (b)	3	Mod 5 DNA and Polypeptide Synthesis	12-6, 12-12
32	7	Mod 8 Epidemiology	12-2, 12-5, 12-15
33(a)	3	Mod 5 Cell Replication	12-5, 12-12
33 (b)	2	Mod 5 Reproduction	12-12
33 (c)	4	Mod 6 Mutation	12-5, 12-13
34 (a)	2	Mod 5 Inheritance Patterns in a Population	12-5, 12-12
34 (b)	4	Mod 6 Mutation	12-13